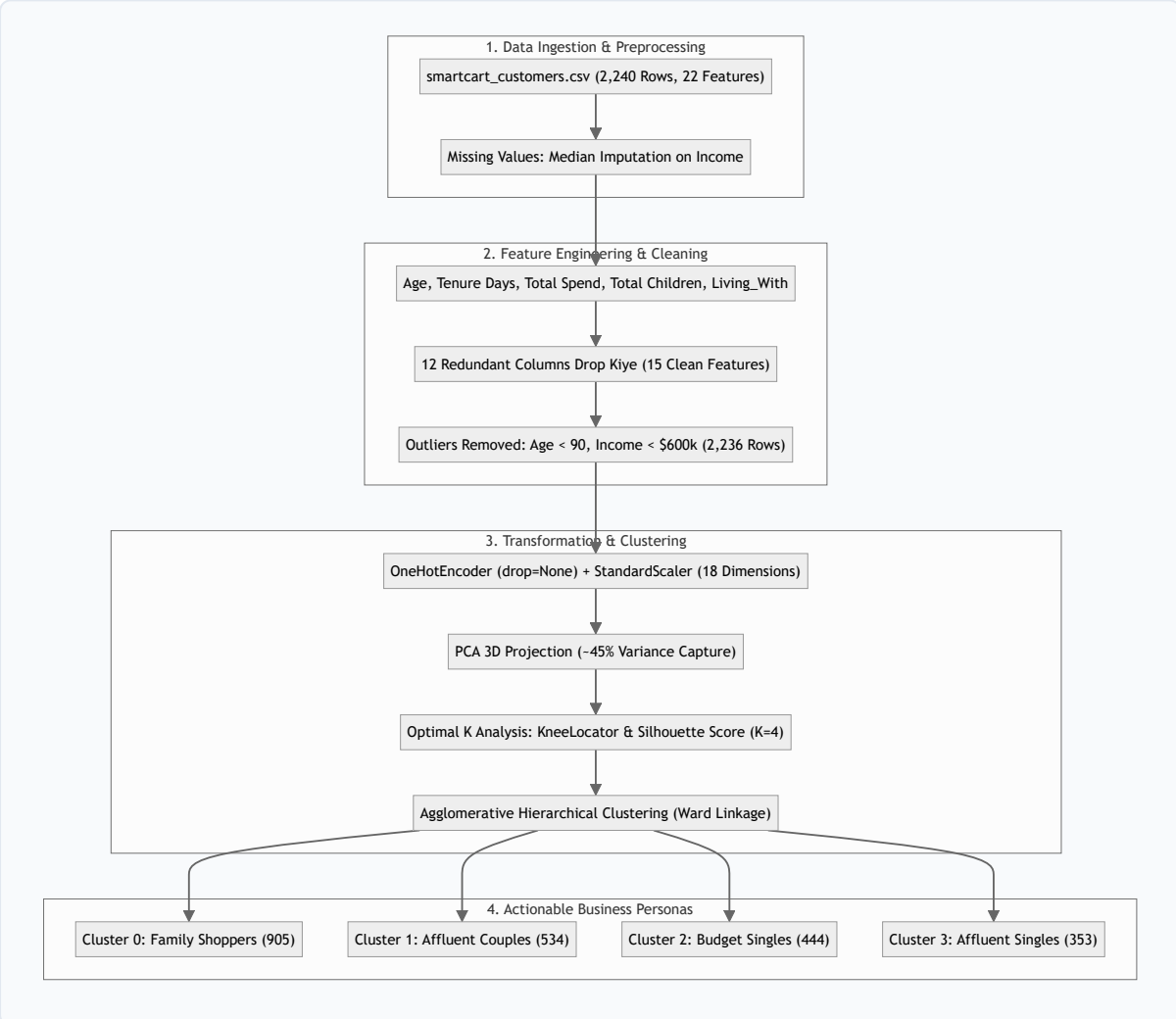
Lecture 33: SmartCart Clustering System (End-to-End Minor Project) (Hinglish)

Course Module Overview

Lecture 33 mein humne **Unsupervised Machine Learning** ka sabse important practical commercial application — **Customer Segmentation System** — end-to-end design aur build kiya hai.

Raw transaction dataset (`smartcart_customers.csv`) se lekar feature engineering, PCA 3D visualization, hyperparameter tuning, Agglomerative Hierarchical Clustering, aur **4 distinct customer personas** ke commercial marketing playbooks tak sabhi steps systematically cover kiye gaye hain.



Submodule Architecture & Index

Submodule	Topic & Coverage	Notebook Code Reference
Notes 33.01	Project Intro, Business Problem & Dataset Audit	Cells 00 - 04 in `smartcart.ipynb`
Notes 33.02	Missing Values Handling: Income par Median Imputation	Cells 05 - 08 in `smartcart.ipynb`
Notes 33.03	Feature Engineering: Age, Tenure, Total Spend, Children, Living_With	Cells 09 - 15 in `smartcart.ipynb`
Notes 33.04	Redundant Columns Drop karna & Curse of Dimensionality	Cells 16 - 20 in `smartcart.ipynb`
Notes 33.05	Outlier Detection via Pairplots & Filtering Bounds	Cells 21 - 23 in `smartcart.ipynb`
Notes 33.06	Correlation Heatmap Analysis & Feature Insights	Cells 24 - 28 in `smartcart.ipynb`
Notes 33.07	OneHotEncoder (bina drop first ke) & StandardScaler	Cells 29 - 39 in `smartcart.ipynb`
Notes 33.08	High-Dimensional Data Visualization: PCA 2D vs 3D	Cells 40 - 45 in `smartcart.ipynb`
Notes 33.09	Optimal K Selection: Elbow Method (KneeLocator) & Silhouette Score	Cells 46 - 53 in `smartcart.ipynb`
Notes 33.10	Clustering: K-Means vs Agglomerative Hierarchical Clustering	Cells 54 - 59 in `smartcart.ipynb`
Notes 33.11	Cluster Characterization: Spending vs Income Patterns	Cells 60 - 64 in `smartcart.ipynb`
Notes 33.12	Cluster Summary Table & 4 Customer Persona Playbooks	Cells 65 - 66 in `smartcart.ipynb`
SmartCart Project	Dedicated Project Directory: Notebook, Dataset, Assignment PDF	`smartcart.ipynb`, `smartcart_customers.csv`

Submodule	Topic & Coverage	Notebook Code Reference
Practice Problems	3 Industry Case Studies with Detailed Solutions	`Solutions_HINGLISH.md`

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4 Customer Personas Summary ($N = 2,236$)

- **Cluster 0 (40.5% - 905)**: *"Family Shoppers / Thrifty Couples"* — Moderate income (\$39.7k), moderate spend (\$222), bacchon ki sankhya (1.24), discount coupons seeker.
- **Cluster 1 (23.9% - 534)**: *"Affluent Couples / Established Spenders"* — High income (\$72.8k), high spend (\$1,237), physical store aur catalog shoppers.
- **Cluster 2 (19.9% - 444)**: *"Budget Singles / Window Shoppers"* — Low income (\$37.0k), low spend (\$166), single parents (1.27 kids), high web visits par low conversion.
- **Cluster 3 (15.8% - 353)**: *"Affluent Singles / Golden Cluster"* — High income (\$70.7k), high spend (\$1,190), living alone, **Highest Campaign Response (32.0%)** with maximum marketing ROI.